

Regime-Aware Dynamic Asset Allocation

A Cross-Asset Transformer with Factor-Structured Covariance Forecasting

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Motivation: Asymmetric Correlation

Markowitz portfolio theory reduces asset allocation to two inputs:

$$\mathbf{w}^* = \arg \min_{\mathbf{w}} -\boldsymbol{\mu}^\top \mathbf{w} + \frac{\gamma}{2} \mathbf{w}^\top \boldsymbol{\Sigma} \mathbf{w} \quad \text{s.t. } \mathbf{1}^\top \mathbf{w} = 1.$$

The covariance $\boldsymbol{\Sigma}$ is what drives diversification.

The catch

$\boldsymbol{\Sigma}$ is *not* stationary. Stock correlations **rise sharply during downturns** — precisely when diversification matters most.

Asymmetric correlation (Longin & Solnik, 2001):

- Pairwise correlations among US equities: ~ 0.3 in calm regimes
- ~ 0.8 in crisis regimes (e.g., 2008, March 2020)

A portfolio diversified under the calm-regime $\boldsymbol{\Sigma}$ is **undiversified** the moment a regime shifts.

What Classical Σ Estimators Miss

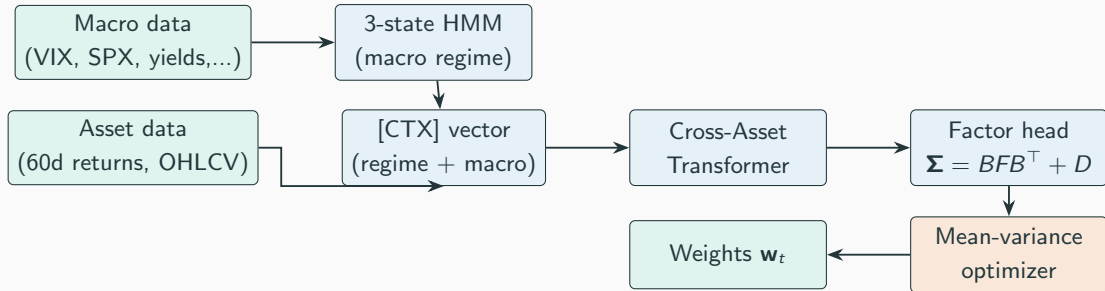
- **Sample covariance** (rolling window): backward-looking, smooth. Catches up to a regime change weeks late.
- **Ledoit–Wolf shrinkage**: stabilizes the estimate but offers no regime-conditional structure.
- **DCC-GARCH**: time-varying, but parametric and slow to adapt.
- **Risk-parity / inverse-vol**: dispenses with μ and Σ jointly; uses only marginal vols.

None of these condition Σ on the current macroeconomic state.

Project goal

Build a Σ estimator that is **conditional on macro regime**, learned from data, and produces a **richer covariance signal** than any of the classical alternatives — then verify it improves out-of-sample risk-adjusted return.

Pipeline: Macro $\rightarrow \Sigma_t \rightarrow$ Weights



Three learned components: HMM (fit by EM), Transformer (trained by SGD on Gaussian NLL), and an explicit convex optimizer.

Data and Universe

Universe: 16 US mega-cap technology equities

AAPL, MSFT, GOOGL, AMZN, META (FB), NFLX, NVDA, TSLA,
ADBE, CRM, ORCL, INTC, CSCO, QCOM, IBM, TXN

Data

- Daily OHLCV from Kaggle [jacksoncrow/stock-market-dataset](#) (through Apr 2020), extended to Dec 2024 via [yfinance](#).
- Macro: VIX, SPX, 3m/10y/30y Treasury yields, DXY, crude oil, HYG/TLT credit proxy.
- Total time series: **5,738 trading days** (2003-01-02 → 2024-12-30).
- Availability mask: TSLA enters 2010-06, META 2012-05 (universe grows 14 → 16).

Backtest window

- 2008-01-02 through 2024-12-30 (17 years), **4,417 out-of-sample days**.
- **897 weekly rebalance decisions** (horizon $h = 5$ trading days).
- **18 walk-forward folds**, each test window = 252 days ≈ 1 year.

Model Inputs: One [CTX] Token + 16 Asset Tokens

Asset token (66-d, per stock):

- 60 trailing daily log returns
- rolling 60-day realized volatility
- previous portfolio weight $w_{t-1,i}$
- 4 OHLCV-derived features:
 - intraday range $(H-L)/C$
 - intraday return $(C-O)/O$
 - overnight gap $(O-C_{\text{prev}})/C_{\text{prev}}$
 - volume z-score

[CTX] token (16-d, shared):

- 3 HMM regime posteriors (calm / mid / stress)
- VIX level + 60-day log change
- SPX 60-day return
- 10y Treasury yield + 60-day change
- DXY 60-day return
- Crude oil 60-day return
- HYG/TLT credit spread + change
- 10y–3m & 30y–10y term spreads + changes

Each token projects through a separate linear layer into a **shared 192-d model space**; the 17 vectors then enter a 5-layer transformer.

Hidden Markov Macro Regime Detector

Observation per day: 2-d vector $X_t = (\text{SPX log return}, \log \text{VIX})$

Model: 3-state Gaussian HMM, full covariance per state.

$$P(X_t \mid s_t = k) = \mathcal{N}(X_t \mid \mu_k, \Sigma_k), \quad P(s_t \mid s_{t-1}) = A_{s_{t-1}, s_t}.$$

Fit by Baum-Welch (EM); **outside the gradient graph** of the transformer.

- State 0 = lowest mean log-VIX (**calm**)
- State 1 = middle (**mid**)
- State 2 = highest (**stress**)
- Post-fit relabeling by mean log-VIX gives stable semantics across re-fits.

Output: 3-vector $[P(\text{calm}_t), P(\text{mid}_t), P(\text{stress}_t)]$, fed into the [CTX] token.

Sequence layout: 1 [CTX] + 16 asset tokens $\Rightarrow$ length-17 sequence in 192-d.

- **Two projections:** $W_{[\text{CTX}]} \in \mathbb{R}^{192 \times 16}$ for the context, $W_{\text{asset}} \in \mathbb{R}^{192 \times 66}$ **shared across all 16 assets**.
- Asset identity carried by **learned positional embeddings** $p_1, \dots, p_{16}$.
- 5 pre-LN transformer blocks; 8 heads; FF expansion $\times 4$; dropout 0.15.
- Self-attention lets every asset attend to every other asset **and** to the [CTX] row $\Rightarrow$ cross-asset interactions are dynamic, input-dependent.
- **Mask:** inactive assets (e.g., TSLA pre-2010) excluded at every attention layer.

Total parameters $\approx 1.8 \times 10^6$, all trained jointly under one loss.

Factor-Structured Covariance Head

A full Cholesky parametrization would predict $N(N+1)/2 = 136$ entries per snapshot — too many for ~ 500 – $2,500$ training points per fold.

Rank-4 factor structure:

$$\Sigma_t = \mathbf{B}_t \mathbf{F}_t \mathbf{B}_t^\top + \mathbf{D}_t$$

- $\mathbf{B}_t \in \mathbb{R}^{16 \times 4}$ (factor loadings): linear head on each asset row.
- $\mathbf{D}_t = \text{diag}(d_{t,i}^2)$: scalar head + softplus $\rightarrow$ positive idiosyncratic variance.
- $\mathbf{F}_t \in \mathbb{R}^{4 \times 4}$: linear head on the **[CTX]** row, parameterized as $\mathbf{F}_t^{1/2}(\mathbf{F}_t^{1/2})^\top$.

Why this matters: $\mathbf{F}_t$ is the only component derived from the macro/[CTX] state. The factor structure is therefore **explicitly regime-conditioned** — this is where asymmetric correlation enters the model.

Training Objective: Gaussian NLL

For each snapshot, predict $(\boldsymbol{\mu}_t, \boldsymbol{\Sigma}_t)$ and score against the realized 5-day forward return $\mathbf{r}_{t,t+h}$:

$$\mathcal{L}_t = \frac{1}{2|\mathcal{A}_t|} \left[|\mathcal{A}_t| \log(2\pi) + \log \det \boldsymbol{\Sigma}_t + (\mathbf{r}_{t,t+h} - \boldsymbol{\mu}_t)^\top \boldsymbol{\Sigma}_t^{-1} (\mathbf{r}_{t,t+h} - \boldsymbol{\mu}_t) \right]$$

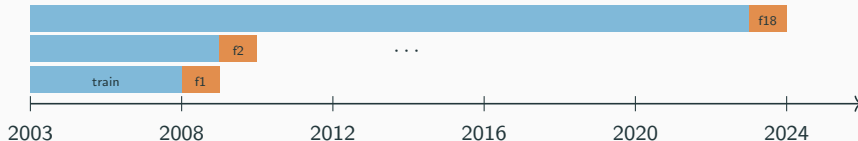
Why this loss?

- **Principled**: maximum likelihood under a multivariate Gaussian.
- **Couples $\boldsymbol{\mu}$ and $\boldsymbol{\Sigma}$** : both heads train under one loss; the quadratic form $\mathbf{r}^\top \boldsymbol{\Sigma}^{-1} \mathbf{r}$ shrinks residuals, the $\log \det \boldsymbol{\Sigma}$ term prevents the model from inflating $\boldsymbol{\Sigma}$ arbitrarily.
- **Per-active-asset normalization** stabilizes loss magnitude as the universe grows ($14 \rightarrow 16$).
- **Cholesky parameterization** of $\boldsymbol{\Sigma}$ makes $\log \det$ and $\boldsymbol{\Sigma}^{-1}$ computable in $O(N^2)$ via triangular solves.

Optimizer: AdamW, $\eta = 3 \times 10^{-4}$, weight decay 5×10^{-4} , gradient clip 1.0, cosine annealing, 6-epoch early-stopping patience.

Walk-Forward Training + Warm-Start Fine-Tuning

18 expanding-window folds; fold f trains on $[t_{\text{start}}, t_f]$, tests on the next 252 days.



Per-fold internal split: last 15% of train period serves as **validation**; best-validation weights restored at fold end. **Warm-start:**

- Fold 0: random init, ~ 30 epochs.
- Fold $f \geq 1$: initialize from fold $(f-1)$ weights, LR scaled by $\eta_{\text{warm}} = 0.5$.
- HMM is **re-fit per fold** (not warm-started) using only data up to the fold's training cutoff.
- Ablation: cold-init each fold collapses Sharpe from $1.02 \rightarrow 0.27$.

Inference Pipeline

At each weekly rebalance date t :

1. Update HMM posteriors $\rightarrow$ [CTX] vector.
2. Forward pass through transformer $\rightarrow (\boldsymbol{\mu}_t^{\text{model}}, \boldsymbol{\Sigma}_t^{\text{model}})$.
3. **Shrinkage** (Ledoit-Wolf-style):

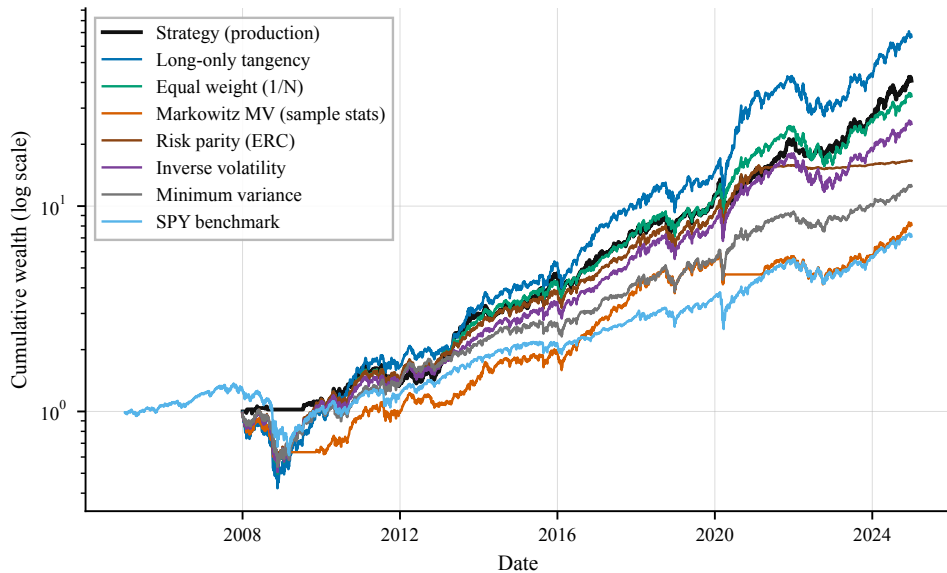
$$\boldsymbol{\Sigma}_t^{\text{final}} = \alpha \boldsymbol{\Sigma}_t^{\text{model}} + (1 - \alpha) \boldsymbol{\Sigma}_t^{\text{sample}}, \quad \alpha = 0.7$$

4. Replace learned $\boldsymbol{\mu}$ with a **60-day sample mean** (justified by ablation).
5. Solve the convex program with Clarabel via CVXPY:

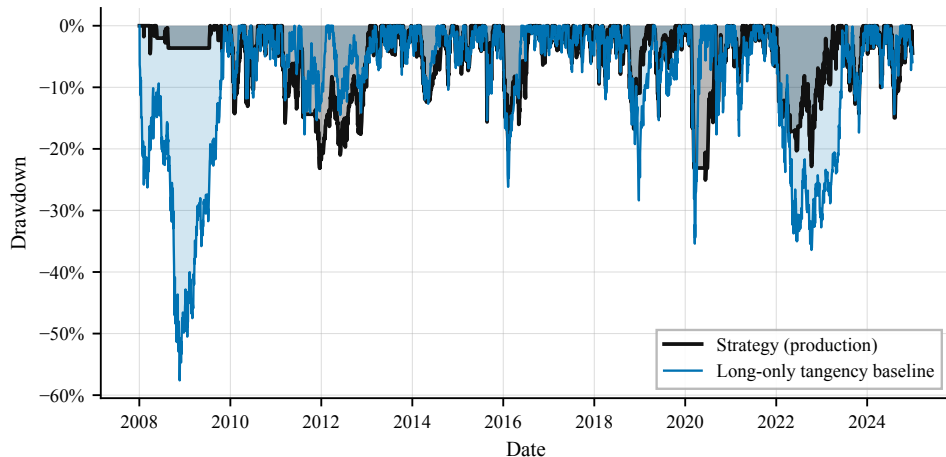
$$\begin{aligned} \mathbf{w}_t^* &= \arg \min_{\mathbf{w}} -\boldsymbol{\mu}_t^\top \mathbf{w} + \frac{\gamma}{2} \mathbf{w}^\top \boldsymbol{\Sigma}_t^{\text{final}} \mathbf{w} + \lambda_{\text{TO}} \|\mathbf{w} - \mathbf{w}_{t-1}\|_1 \\ \text{s.t. } \quad &\mathbf{1}^\top \mathbf{w} = 1, \quad 0 \leq w_i \leq 0.20, \quad \sqrt{\mathbf{w}^\top \boldsymbol{\Sigma}_t^{\text{final}} \mathbf{w}} \leq 0.040. \end{aligned}$$

The weekly volatility cap is the explicit risk-control lever that converts “good $\boldsymbol{\Sigma}$ forecast” into “small drawdown.”

Results: Out-of-Sample Equity Curves



Results: Drawdown vs. Long-Only Tangency Baseline



The tangency baseline finishes at higher cumulative wealth, but at -57.6% maximum drawdown (2008-11-20). Our strategy's MaxDD is -25.1% (2020-06-11), and only -4.5% across the 2008 GFC — the regime-conditional Σ tells the optimizer to step out when correlations spike

Results: Risk-Adjusted Performance Ranking

Strategy	CAGR	Sharpe	MaxDD	Calmar
Production (ours)	23.6%	1.21	−25.1%	0.94
Long-only tangency	27.0%	1.02	−57.6%	0.47
Equal-weight (1/ <i>N</i>)	22.3%	0.96	−51.4%	0.43
Inverse-volatility	20.1%	0.91	−50.1%	0.40
Risk-parity (ERC)	17.3%	0.87	−49.9%	0.35
Minimum-variance	15.4%	0.83	−47.5%	0.32
Sample-stats MV	12.6%	0.71	−38.9%	0.32
SPY benchmark	10.3%	0.61	−55.2%	0.19

Best Sharpe, best Calmar, smallest drawdown among all 7 baselines + SPY.

Beta vs. SPY: 0.51; alpha (annualized): 17.6%.

Negative Results: 18 Attempts, 18 Regressions

We tried 18 architecture/optimizer modifications. **All reduced out-of-sample Sharpe.**

Variant	Δ Sharpe vs. 1.21
12-1 Jegadeesh-Titman momentum μ	-0.41
Scenario-based CVaR optimizer	-0.36
Long/short relaxation ($w_i \in [-0.1, 0.2]$)	-0.29
5-seed ensemble	-0.18
Realized-covariance auxiliary loss	-0.14
Cold-init each fold (no warm-start)	-0.94
Larger transformer ($d_{\text{model}} = 320$)	-0.15
Daily training stride (v17b)	-0.22
GNN over asset interactions	-0.31
<i>... and nine more, all negative</i>	

Interpretation: the v7b config is a tight, opinionated local optimum. Its success comes from **constrained, low-DOF architecture**, not from flexibility.

Conclusion

What we built

- Macro-regime-conditional covariance forecaster: HMM + cross-asset Transformer + factor head.
- Trained under Gaussian NLL on 5-day forward returns; rigorous 18-fold walk-forward protocol.

What we found

- **Sharpe 1.21, Calmar 0.94, MaxDD -25.1%** OOS over 2008–2024.
- Beats 7 classical baselines + SPY on every risk-adjusted metric.
- Architecture is brittle to perturbation — 18 of 18 ablations regressed.

Limitations

- Universe selected from a 2020 snapshot → survivorship bias.
- No transaction-cost model; gross-of-cost Sharpe.
- Single asset class (equities). Multi-asset extension untested.